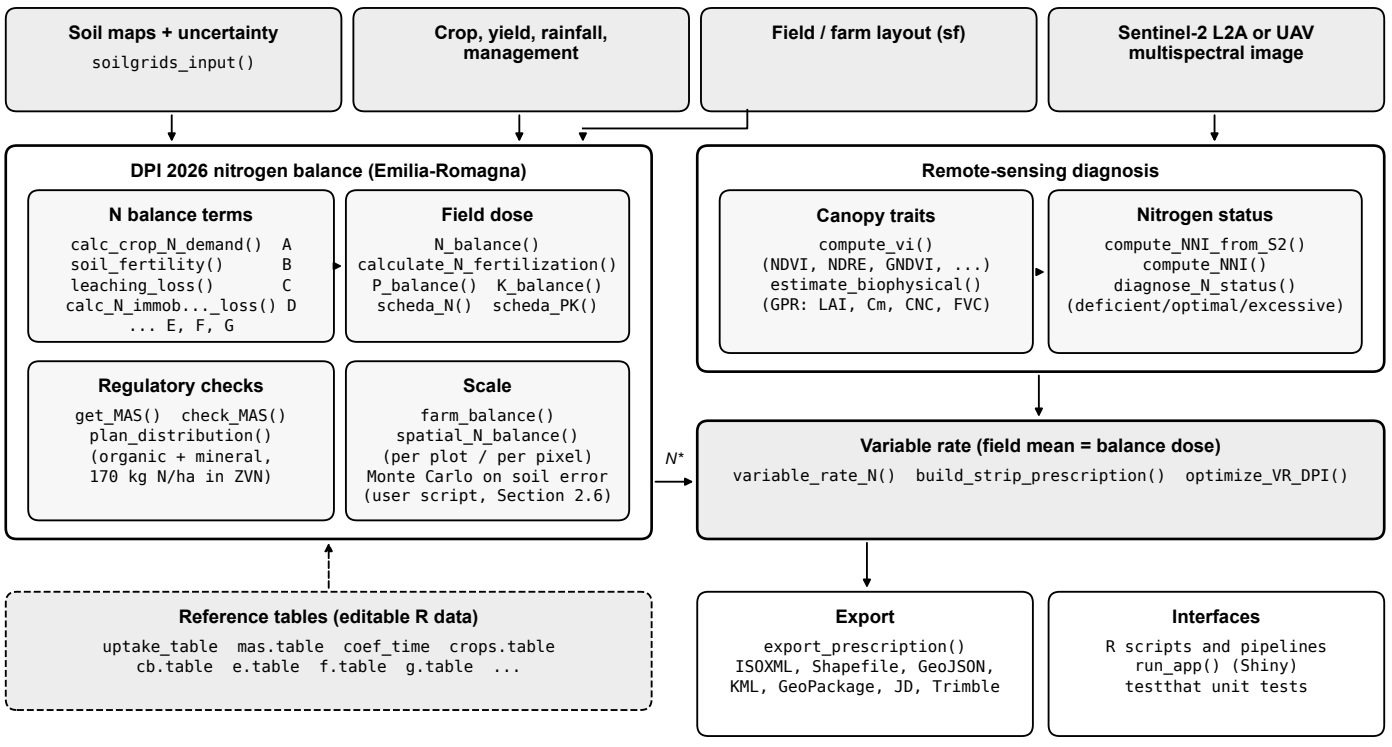
Inputs
Soil maps + uncertainty
soilgrids_input()
Crop, yield, rainfall, management
Field / farm layout (sf)
Sentinel-2 L2A or UAV multispectral image
DPI 2026 nitrogen balance (Emilia-Romagna)
N balance terms
calc_crop_N_demand() A
soil_fertility() B
leaching_loss() C
calc_N_immob..._loss() D
... E, F, G
Field dose
N_balance()
calculate_N_fertilization()
P_balance() K_balance()
scheda_N() scheda_PK()
Regulatory checks
get_MAS() check_MAS()
plan_distribution()
(organic + mineral,
170 kg N/ha in ZVN)
Scale
farm_balance()
spatial_N_balance()
(per plot / per pixel)
Monte Carlo on soil error
(user script, Section 2.6)
Remote-sensing diagnosis
Canopy traits
compute_vi()
(NDVI, NDRE, GNDVI, ...)
estimate_biophysical()
(GPR: LAI, Cm, CNC, FVC)
Nitrogen status
compute_NNI_from_S2()
compute_NNI()
diagnose_N_status()
(deficient/optimal/excessive)
N*
Variable rate (field mean = balance dose)
variable_rate_N() build_strip_prescription() optimize_VR_DPI()
Reference tables (editable R data)
uptake_table mas.table coef_time crops.table
cb.table e.table f.table g.table ...
Export
export_prescription()
ISOXML, Shapefile, GeoJSON,
KML, GeoPackage, JD, Trimble
Interfaces
R scripts and pipelines
run_app() (Shiny)
testthat unit tests